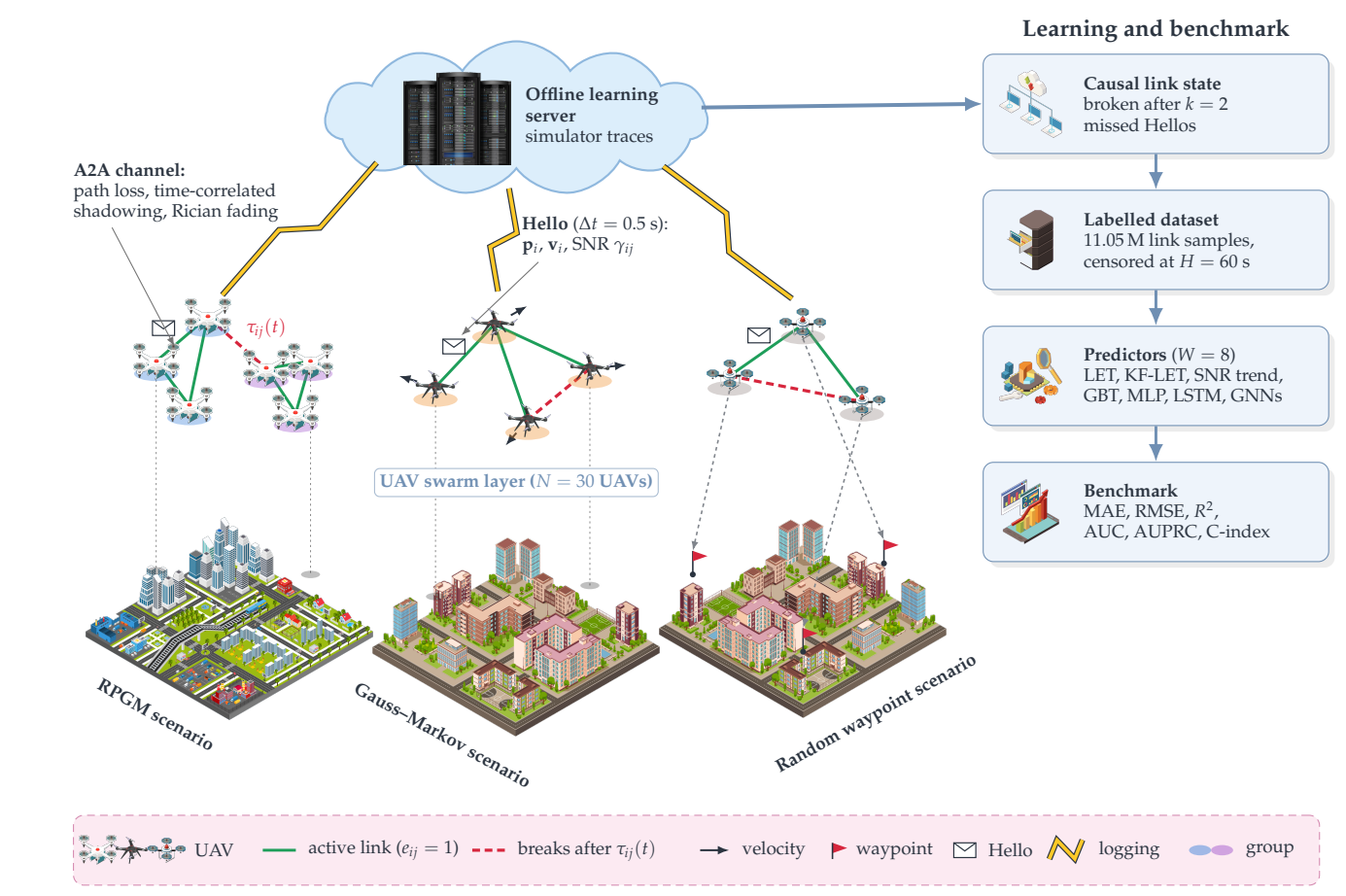

Learning and benchmark
Offline learning server
simulator traces
Causal link state
broken after $k = 2$
missed Hellos
Labelled dataset
11.05 M link samples,
censored at $H = 60$ s
Predictors ($W = 8$)
LET, KF-LET, SNR trend,
GBT, MLP, LSTM, GNNs
Benchmark
MAE, RMSE, R^2,
AUC, AUPRC, C-index
A2A channel:
path loss, time-correlated
shadowing, Rician fading
Hello ($\Delta t = 0.5$ s):
$\mathbf{p}_i$, $\mathbf{v}_i$, SNR γ_{ij}
$\tau_{ij}(t)$
UAV swarm layer ($N = 30$ UAVs)
RPGM scenario
Gauss–Markov scenario
Random waypoint scenario
UAV
active link ($e_{ij} = 1$)
breaks after $\tau_{ij}(t)$
velocity
waypoint
Hello
logging
group